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5 May 1986
SUPERSEDING
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MILITARY SPECIFICATION

DISINFECTANT, FOOD SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This document covers one type and grade of disinfectant, in powder form.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366	- Sieve, Test
PPP-B-26	- Bag, Plastic (General Purpose)
PPP-B-566	- Boxes, Folding, Paperboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6840

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PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup

MILITARY

MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

ENVIRONMENTAL PROTECTION AGENCY

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act

(The code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Reprints of certain regulations and application forms for registration of economic poisons may be obtained from the Pesticides Regulations Division, Environmental Protection Agency, 401 M Street SW, Washington, DC 20460.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

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SOAP AND DETERGENT ASSOCIATION

ABS and LAS Biodegradability Scientific and Technical Report No. 3

(Application for copies should be addressed to the Soap and Detergent Association, 475 Park Avenue, New York, NY 10016.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeia Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified at the time a bid is submitted, the contractor shall submit to the contracting officer a certified copy of a current year laboratory report covering the quality and grade of materials that he proposes to deliver. The laboratory report shall contain test data demonstrating that the food service disinfectant proposed for delivery by the contractor has been tested in accordance with the tests in section 4 and found to comply with the requirements of this document. Any one of the following types of reports will be satisfactory for this requirement:

a. An independent or commercial laboratory report.
b. The prospective contractor's own laboratory report.
c. A Governmental laboratory report from a contract.

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The purpose of this requirement is to assist the Government in determining the capability of contractors to conform to all the requirements of this document. The submission of an acceptable report under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of this document on all deliveries.

3.2 Material. The food service disinfectant shall consist only of the following materials, as applicable: trichloromelamine, a surface active material, anhydrous citric acid, anhydrous monosodium dihydrogen phosphate, and potassium iodide (see 6.4).

3.2.1 Surface active material. The surface active material shall be white to a light cream in color, anionic type, and the active ingredient of which consists of linear sodium alkyl (C_{11} - C_{12}) aryl sulfonate. The surface active material shall be at least 80 percent biodegradable when tested as specified in 4.3.1.1.

3.2.1.1 Iodine uptake. The iodine uptake shall be no more than 4.5 milligrams (mg) of iodine per gram of surface active material when tested as specified 4.3.1.1.

3.2.1.2 Active ingredient. The surface active material shall have an active ingredient content of not less than 81 percent when tested as specified in 4.3.1.1. (Commercially available surfactants suitable for use in this item are generally 81 to 87 percent active.)

3.2.1.3 PTH factor (p-toluidine hydrochloride). The surface active material shall have a PTH factor of 338 to 351 when tested as specified in 4.3.1.1.

3.2.1.4 Moisture content. The moisture content shall be no greater than 1.25 percent when tested as specified in 4.3.1.1.

3.2.2 Potassium iodide. The potassium iodide shall be in granular form and shall conform to the requirements of the United States Pharmacopeia.

3.3 Composition. When tested as specified in 4.3.2.5, the food service disinfectant shall conform to the requirements of the standard trichloromelamine formula specified in table I. In addition, the disinfectant requires the addition of a potassium iodide pouch, packaged separately, to the standard trichloromelamine pouch.

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TABLE I. Standard trichloromelamine formula

Ingredient	Percent by weight Minimum	Maximum
Trichloromelamine	26.9	27.9
Anionic linear sodium alkyl aryl sulfonate surface active material, technical (LAS)	11.5	12.5
Citric acid, anhydrous	51.0	55.0
Monosodium dihydrogen phosphate, anhydrous	Remainder	

* 3.3.1 4.77 ounces trichloromelamine-potassium iodide pouches. Unless otherwise specified (see 6.2), each trichloromelamine pouch shall contain 3.43 ± 0.10 ounces of the standard trichloromelamine formulation and each potassium iodide pouch shall contain 1.34 ± 0.10 ounces of potassium iodide. The pouches, prepared as separate units, shall be constructed of heat sealable barrier material consisting of the following lamination of materials (from inside to outside surface): 0.0015-inch polyethylene by 0.002-inch aluminum foil by 0.0015-inch polyethylene by 25-pound basis weight pouch paper with 2-pound basis weight pouch paper with 2-pound basis weight per ream (24 by 36-500) lacquer coating. Pouches shall be one or two piece construction. Side, top, and bottom seams of the two-piece pouch shall be heat sealed with the seals a minimum of 3/16 inch in width. In addition to a minimum of 3/16 inch seal on the top of each pouch, a lip of at least 1/2 inch shall be present at the top of each pouch for joining and heat sealing both the trichloromelamine and the potassium iodide pouches together. After each pouch has been filled and closure heat sealed, one trichloromelamine pouch labeled pouch "A" and one potassium iodide pouch labeled pouch "B" shall be heat sealed together by joining the top 1/2-inch lip of each pouch. In addition, a tearing notch shall be positioned in top seam of each trichloromelamine and potassium iodide pouch. The notch in each pouch shall be located in the top seam of each pouch so that both pouches may be opened in one operation to ensure delivery of the contents of both pouches during use. One-piece pouch seams and closure shall be as specified in 3.3.2 and 3.3.3. When tested as specified in 4.4.3.1 and 4.4.3.2, the finished pouches shall show no signs of visual deterioration. The exterior of the pouches shall be free from visible

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dust. The joined pouches shall be enclosed in a close-fitting heat sealed bag, fabricated from material conforming to type II, grade A, class 1 of MIL-B-121.

3.3.2 Pouch side seams and closure. All side seams and closures shall be heat sealed. The seal shall be a minimum of 3/16 inch wide.

3.3.3 Bottom fold sealing. The bottom fold-over shall be heat sealed. The seal shall be a minimum of 1/16 inch wide.

3.3.4 Leakage of seams and closure of pouch. The seams and closure of the pouch shall be complete and continuous and shall not show signs of leakage or bubbling when immersed in water, or shall not wet the contents when tested as specified in 4.4.3.3.

3.4 Trichloromelamine pouch, finished product.

3.4.1 Titratable chlorine. The contents of the trichloromelamine pouch shall contain not less than 24.0 percent nor more than 26.5 percent of titratable chlorine when tested as specified in 4.3.2.5.

3.4.2 Acidity. The contents of the trichloromelamine pouch shall have a pH value not less than 2.5 and shall have a titratable acidity equivalent to not less than 200 mg of sodium hydroxide per gram of product when tested as specified in 4.3.2.5.

3.4.3 Surface active ingredient. The contents of the trichloromelamine pouch shall have not less than 9.1 percent nor more than 10.7 percent of active ingredient when tested as specified in 4.3.2.5. (These values are slightly less than theoretical for there is seldom complete conversion in the test method in 4.4.2.3.)

3.4.4 Solubility. The contents of the trichloromelamine pouch shall be soluble when tested as specified in 4.3.2.5. The titratable chlorine in the sample shall be not less than 152 mg per liter.

3.4.5 Moisture. The content of the trichloromelamine pouch shall have a moisture content of not more than 0.60 percent when tested as specified in 4.3.2.5.

3.4.6 Particle size. Not less than 95 percent, by weight, of the contents of the trichloromelamine pouch shall pass through a No. 20 standard sieve conforming to RR-S-366, and not less than 100 percent through a No. 10 standard sieve conforming to RR-S-366, when tested as specified in 4.3.2.5.

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3.4.7 Bactericidal performance. When tested as specified in 4.3.2.5, the contents of the trichloromelamine pouch shall give a 99.999 percent kill in 90 seconds for at least 2 of the 3 tests.

3.4.8 Formulation stability. The contents of the trichloromelamine pouch shall not lose more than 20 percent of the original amount of titratable chlorine when determined as specified in 4.3.2.5.

3.4.9 Surface tension. A solution of the contents of the trichloromelamine pouch shall have a surface tension of not more than 40 dynes per centimeter when prepared and tested as specified in 4.3.2.5.

* 3.5 Labeling. The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act and in compliance with its requirements. The contractor shall use EPA Registration No. 40510-1 assigned to the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014. The contractor shall use the EPA Establishment number assigned to his manufacturing plant. The EPA approved labels shall be as shown on figures 1 and 2. In addition to the marking specified in 5.4, the information found on figures 1 and 2 shall be legibly marked on each bag (containing the double compartmented pouch) and each interior box.

3.5.1 Additional marking. In addition to the above markings, the following information shall be legibly printed on both sides of each appropriate pouch:

Trichloromelamine Pouch A.
Potassium Iodide Pouch B.

3.5.2 Instruction sheet. A sheet of chemical wood writing paper, legibly printed with the following instructions, shall be inserted in each interior box specified in 5.1.1:

INSTRUCTIONS FOR DETERMINATION OF
TITRATABLE CHLORINE WHEN DISINFECTANT,
FOOD SERVICE IS TO BE ISSUED
AFTER _____ (the same date stamped on caution label affixed to shipping container shall be inserted here).

Weigh out 1 gram of the contents of the trichloromelamine pouch to the nearest milligram into a round bottom test tube (approximately 14 millimeters (mm) in diameter and 160 mm in length). Add 0.5 milliliter (mL) of distilled water to the sample, which is then stirred with a rounded tipped stirring rod (of a size convenient to fit into tube). Mix the sample into a smooth paste for

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exactly 3 minutes. Completely transfer the paste by rinsing into a 1.5 liter beaker, using a total of exactly 1 liter of distilled water. Stir the solution for exactly 30 seconds with a large stirring rod (do not cause whipping or splashing when stirring) and then let stand for 2 minutes. At the end of this time, insert a 20 mL pipette 1 inch below the surface and draw 20 mL into the pipette. Place the 20 mL sample into a 125 mL Erlenmeyer flask or iodine flask and analyze for titratable chlorine by the following procedure:

Dissolve 1 gram of potassium iodide crystals in the 20 mL sample. Add six drops of 50 percent acetic acid and swirl to ensure complete mixing. Allow to stand five minutes in the dark. Titrate the free iodine at once with freshly standardized 0.005 N sodium thiosulfate. When the color is straw yellow, add starch indicator and titrate to end point. Analysis should be conducted in triplicate and results reported as an average.

Calculation:

$$\text{Percent titratable chlorine} = \frac{(35.45)\ (\text{mL of sodium thiosulfate})\ (N)\ (50)\ (100)}{(\text{gram sample})\ (1000)}$$

3.6 Workmanship. The food service disinfectant shall be a noncaking, clean, uniform powder. The entire process of mixing, grinding (see 6.3 and 6.5), filling, and sealing of the individual pouches shall be carried out in surroundings maintained at a relative humidity of not more than 45 percent and at a temperature of not more than 80°F, and in such a manner as to ensure a product of consistently high quality. All components and bulk mixed finished formulation shall be unit packed in moisture-vapor proof containers and kept dry and protected from moisture until incorporated into final packaging.

3.6.1 Additional workmanship. The potassium iodide cannot be mixed intimately with the rest of the trichloromelamine formulation. The unit packing of the potassium iodide and the trichloromelamine finished product shall be carried out in separate rooms under the controlled humidity and temperature conditions mentioned above. A two-compartmented pouch type package shall be used to keep the potassium iodide separated from the trichloromelamine formulation until time for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Pre-acceptance inspection.

4.2.1 In-process. In-process examination of the batch mix shall be made to determine that composition formulation complies with 3.3 and that the workmanship complies with 3.6.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Inspection of materials and components. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, or qualified in this document or applicable purchase document.

4.3.1.1 Testing of components. Component testing shall be performed in accordance with table II. Each lot of surface active material shall be checked for all requirements prior to incorporation into formulation. For purposes of sampling, a lot shall consist of all the surface active material received from one manufacturer, produced at one plant from the same raw materials and under essentially the same manufacturing conditions. The lot size shall be expressed in pounds of surface active material. The sample unit for testing, except color, shall be a one-half pound composite obtained by randomly selecting and thoroughly mixing one-quarter pound samples in the frequency indicated below. The composite shall be placed in a clean, dry, sealed container of the smallest possible size. Care shall be exercised to

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prevent contamination or alteration of the composite during the sampling, weighing, and storage. The lot shall be unacceptable if the composite fails to meet any of the test requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (pounds)	Frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II. Instructions for testing of components

Characteristic	Document reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind unit	Lot avg.		Pass or fail	Numerically to nearest
Surface active material							
Color	3.2.1	4.3.1.1	X 1/			X	
Biodegradability	3.2.1	4.4.1.1		X	2		Average of 2 determinations to nearest 0.1 percent
Iodine uptake	3.2.1.1	4.4.1.2		X	2		Average of 2 determinations to nearest 0.5 mg
Active ingredient	3.2.1.2	4.4.1.3		X	2		Average of 2 determinations to nearest percent
PTH factor	3.2.1.3	4.4.1.4		X	2		Average of 2 determinations to nearest number

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TABLE II. Instructions for testing of components - Continued

Characteristic	Document Reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind unit	Lot avg.		Pass or fail	Numerically to nearest
Moisture	3.2.1.4	4.4.1.5		X	2		Average of 2 determinations to nearest 0.01 percent
Potassium iodide	3.2.2	2/					

1/ All containers shall be examined for color. Failure of the contents of a container to meet the color requirement shall be cause for rejection of the contents of that container.

2/ The container shall furnish a certificate of compliance covering this requirement.

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4.3.2 Inspection of the end item. The disinfectant and pouches shall be examined for the defects in the applicable subparagraphs. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be the number of packages filled from a single homogeneous blend of the formulated product and shall be expressed as indicated in the applicable examination paragraphs.

* 4.3.2.1 Examination of the filled heat sealed bags and the double compartmented pouches for defects in construction, workmanship, and markings. The heat sealed bags and the filled pouches shall be examined for defects in construction, workmanship, and markings as listed in table III. The lot size for this examination shall be expressed in units of filled heat sealed bags. The sample unit for this examination shall be one filled heat sealed bag. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

TABLE III. Examination of filled heat sealed bags and double compartmented pouches for construction defects

Examine	Defects
Construction	Outer bag not constructed of materials specified Outer bag not sealed Trichloromelamine pouch missing Potassium iodide pouch missing Pouches not joined at the top Tearing notch missing from pouches Tearing notch not positioned as specified Not in accordance with contract requirements Seams and closures of pouches not completely heat sealed throughout entire length and width Not uniformly constructed Pouches not constructed of materials specified Any evidence of seepage; any unsound seam Any tear, cut, slit split, cracks, blisters, or holes (including pinholes)

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TABLE III. Examination of filled heat sealed bags and double compartmented pouches for construction defects - Continued

Examine	Defects
Workmanship	Not clean; any evidence of dirt, dust, or foreign matter Presence of food service disinfectant, or potassium iodide on exterior of pouches, as applicable Caking; not uniform powder
Markings	Markings pertaining to the applicable information and instructions for end item use, missing, illegible, incomplete, or does not conform to applicable legend specified in 3.5 Not printed in black or blue ink Printing ink faded; blurred; smeared

4.3.2.2 Examination of the double compartmented pouches for dimensional defects. The filled bags shall be examined for the defects in table IV. The lot size for this examination shall be the number of heat sealed bags filled from a single homogeneous blend of the formulated product and shall be expressed in units of filled heat sealed bags. The sample unit shall be one filled heat sealed bag. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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TABLE IV. Dimensional defects

Examine	Defects
Size of seams of pouch:	
Two piece construction	Side, top, and bottom seams less than 3/16 inch wide
One piece construction	Side seams and closures less than 3/16 inch wide Bottom fold seal less than 1/16 inch wide Less than 1/2-inch lip present for joining and sealing trichloromelamine pouch to potassium iodide pouch

4.3.2.3 Examination of the filled pouches for net contents. The lot size for this examination shall be expressed in units of filled pouches. The sample unit shall be one filled pouch. The net contents shall be not less than specified or indicated. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

* 4.3.2.4 Examination of intermediate pack. The completed intermediate pack, filled and sealed as specified in 5.1.1, shall be examined for the defects in table V. The lot size for this examination shall be the number of filled and sealed intermediate packs available for inspection at one time. The sample size shall be one filled and sealed intermediate pack. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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TABLE V. Examination of intermediate packs

Examine	Defect
Box	Not properly closed Not correct size
Heat sealed wrap	Heat sealed seam not as specified Excess air within bag

4.3.2.5 Testing of the end item. The methods of testing listed in table VI shall be followed. The lot size shall be expressed in units of bags. The sample unit for testing the trichloromelamine pouches shall be five individual 3.43-ounce pouches. The sample unit for packaging testing shall be one polyethylene bagged intermediate box. The number of sample units to be tested shall be as indicated below. When specified in table VI, tests shall be performed on a sample unit or composite basis. The lot shall be unacceptable if a sample unit or the composite fails to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (bags or pouches)	Number of units
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE VI. Instructions for testing of the end item

Characteristic	Document reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerically to nearest
Composition	Table I	1/					
Trichloromelamine:							
Titratable chlorine	3.4.1	4.4.2.1	X		2		Average of 2 determinations to nearest 0.1 percent
Acidity	3.4.2						
pH value		4.4.2.2	X		1		0.1 pH
Titratable acidity		4.4.2.2	X		1		milligram
Surface active ingredient	3.4.3	4.4.2.3		X	3		Average of 3 determinations to nearest 0.1 percent
Solubility	3.4.4	4.4.2.4		X	1	X	
Moisture	3.4.5	4.4.2.5	X		2		Average of 2 determinations to nearest 0.01 percent
Particle size	3.4.6						
Standard sieve No. 20		4.4.2.6		X	1		0.1 percent/weight
Standard sieve No. 10		4.4.2.6		X	1		0.1 percent/weight

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TABLE VI. Instructions for testing of the end item - Continued

Characteristic	Document reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerically to nearest
Bactericidal performance*	3.4.7	4.4.2.7		X	3		Average of 3 determinations to 0.001 percent kill
Formulation stability	3.4.8	4.4.2.8	X		1	X	
Surface tension	3.4.9	4.4.2.9		X	3		Average of 3 determinations to nearest dyne/cm
Pouches, filled and sealed:							
Finished pouch stability*	3.3.1	4.4.3.1	X		1	X	
Stability after packaging	3.3.1	4.4.3.2	X		1	X	
Leakage of seams and closures	3.3.4	4.4.3.3	X		1	X	
Water spray test (interior package)	5.1.1	4.4.3.4	X		1	X	

1/ Certificate of compliance shall be furnished by the contracting officer for composition requirements except for surface active material.

* Tests performed on one out of every five lots thereafter.

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4.3.2.5.1 Composite sample. One 3.43-ounce pouch shall be selected from each sample unit. The contents of the pouches so selected shall be thoroughly mixed to yield a homogeneous composite. The composite shall be placed in an airtight and watertight brown glass container, covered, and sealed to prevent any moisture change.

4.3.2.5.2 Sample unit basis. The remaining pouches shall be used for testing on a sample unit basis. Care shall be exercised to prevent any change in the compound due to moisture prior to testing.

4.3.2.5.3 Frequency of testing. Except when otherwise specified in table VI, the tests for each characteristic shall be performed on a lot by lot basis. Those tests designated by an asterisk shall be performed on one out of every five lots thereafter.

* 4.3.3 Packaging inspection. An inspection shall be made to determine that the preservation, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with table VII. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed in table VII shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the the AQL shall be 2.5 defects per hundred units.

TABLE VII. Packaging defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete closure of container flap, loose strapping, improper taping, or inadequate strapping Bulged or distorted container
Content	Number of pouches of disinfectant per intermediate pack or shipping container is more or less than specified

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* 4.3.4 Palletization inspection. An inspection shall be made to determine that the palletization complies with section 5 requirements. Defects shall be scored in accordance with table VIII. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1, and the AQL shall be 6.5 defects per hundred units.

TABLE VIII. Palletization defects

Examine	Defect
Finished dimensions	Length, width, or height exceeds maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.4 Test procedures.

4.4.1 Tests on surface active material.

4.4.1.1 Biodegradability. The biodegradability of the surface active material shall be determined in accordance with the ABS and LAS Biodegradability Scientific and Technical Report No. 3.

4.4.1.2 Iodine uptake. Weigh sample of 2 grams of the LAS surface active material to the nearest 0.005 gram on a piece of waxed paper. Transfer it to a 250 mL separatory funnel (Corning No. 6340 or equivalent with stem cut to approximately 0.5 inch length) containing 50 mL of hot distilled water. Without shaking, dilute to 75 mL with hot distilled water and swirl but do not shake to dissolve. Add 2 mL of starch indicator solution and titrate with 0.1 normal (N) iodine solution (12.692 grams iodine and 8 grams potassium iodide per liter of distilled water) until all sulfite is destroyed as indicated by the appearance of the starch-iodine color. Swirl during the titration and

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allow to stand with occassional swirling for 2 minutes after the initial appearance of the starch-iodine color to be certain that the color does not fade. Save titrated solution for 4.4.1.3.

4.4.1.2.1 Calculation.

$$\text{Milligrams iodine} = \frac{(126.92)\ (\text{mL of iodine solution})\ (N)}{(\text{grams of sample})}$$

* 4.4.1.3 Active ingredient. Place a 1.5 gram sample weighed to the nearest milligram in a 250 ml volumetric flask and dilute to volume with distilled water. Mix to dissolve. To a 250 ml stoppered graduate cylinder add 25 mL of dissolved sample solution, 25 mL of methylene blue indicator (10 mL of 0.3 percent methylene blue solution, 8 mL concentrated sulfuric acid, 50 grams R.G. sodium sulfate diluted to one liter with distilled water), 25 mL of chloroform, and 25 mL of distilled water. Titrate with standardized 0.002 N Hyamine 1622 solution in 50 mL buret. Periodically during titration, shake the graduated cylinder 30 times and observe for color changes in layers. Continue titration until color of layers matches. Record number of milliliters required.

* 4.4.1.3.1 Calculation. Calculate the percent active anionic surfactant:

$$\text{Percentage} = \frac{(\text{mL Hyamine 1622})\ (\text{N Hyamine 1622})\ (341)\ (A)}{(10)\ (B)\ (C)}$$

where A = 250 (the sample dilution)
B = the sample weight
C = 25 (aliquot of sample used)

4.4.1.4 Determination of PTH factor (p-toluidine hydrochloride). Proceed as for determination of active ingredient with the following modifications: Sample weight shall be approximately 4 grams. Use 50 mL of p-toluidine hydrochloride solution instead of 25 mL. Draw the carbon tetrachloride extracts into a 100 mL volumetric flask and make up to 100 mL with carbon tetrachloride. Titrate a 25 mL aliquot with 0.1 N NaOH. Dry another 25 mL aliquot to constant weight at 100° to 105°C in a tared beaker. This residue is the p-toluidine salt of the active ingredient.

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4.4.1.4.1 Calculations.

$$\text{Active ingredient} = (\text{grams of p-toluidine salt}) - (0.0850)\ (\text{mL NaOH})\ (\text{N NaOH})$$

$$\text{PTH factor} = \frac{(\text{grams of active ingredient})\ (1000)}{(\text{mL NaOH})\ (\text{N NaOH})}$$

Combining the above:

$$\text{PTH factor} = \frac{(\text{grams of p-toluidine salt})\ (1000)}{(\text{mL NaOH})\ (\text{N NaOH})} - 85.0$$

4.4.1.5 Moisture. Weigh sample of 3 grams of the LAS surface active material to the nearest 0.1 mg in a covered sample dish. Uncover the dish and dry the sample in an oven for 24 hours at 105° ± 2°C. Replace the cover, cool the sample to room temperature in a desiccator, and again weigh the sample to the nearest 0.1 mg.

4.4.1.5.1 Calculation.

$$\text{Percent moisture} = \frac{(\text{loss of weight})\ (100)}{(\text{grams of sample})}$$

4.4.2 Tests on contents of trichloromelamine pouch.

4.4.2.1 Titratable chlorine. Thoroughly and quickly mix approximately 5 grams of the contents of the trichloromelamine pouch, sampled at random, by shaking in a closed glass container. Promptly obtain a representative sample of 1 gram from the 5 gram sample weighed to the nearest milligram, and place in a 500 mL Erlenmeyer flask. To this, add 2 grams of potassium iodide (KI) crystals and approximately 25 mL of glacial acetic acid. Stopper the flask and swirl for about 5 minutes to dissolve the sample. (Note: Carry out determination of titratable chlorine in subdued light.) Add approximately 25 mL of distilled water and again stopper and swirl to ensure complete mixing and solution of KI. Titrate at once with a freshly standardized 0.15 N sodium thiosulfate solution to light straw color, add starch indicator, and continue titration to end point.

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4.4.2.1.1 Calculation.

$$\text{Percent titratable chlorine} = \frac{(35.45)\ (\text{mL of sodium thiosulfate})\ (N)\ (100)}{(\text{grams of sample})\ (1000)}$$

4.4.2.2 Acidity. Weigh 1 gram of the contents of the trichloromelamine pouch to the nearest mg and disperse in 1 liter of distilled water at 25° ± 1 °C. The pH, measured with a glass electrode, shall not be lower than 2.5. Titrate the entire solution with 0.1 N NaOH. The latter solution is added from a burette with stirring, measuring the pH with a glass electrode at suitable increments. As the pH value approaches 5.0, the increments should be made smaller, with pH intervals no greater than 0.2. The titration is continued until a pH of 5.5 has been reached. The number of milliliters of 0.1 N NaOH required to give a pH of 5.0 is then found by interpolation of this data.

4.4.2.2.1 Calculation.

$$\text{Milligrams NaOH per gram} = \frac{(\text{mL of NaOH})\ (N)\ (40)}{(\text{grams of sample})}$$

* 4.4.2.3 Surface active ingredient. Place a 1.5 gram sample weighed to the nearest milligram in a 250 mL volumetric flask and dilute to volume with distilled water. Mix to dissolve. Transfer a 25 mL aliquot into a 125 mL flask. While mixing, add 2.0 N sodium arsenite dropwise to the aliquot. Remove 1 drop of solution with a glass rod and apply to starch-iodide test paper. Continue adding drops until paper no longer turns blue (available chlorine is removed). Discard. Record number of drops.

To a 250 mL stoppered graduate, add 25 mL of dissolved sample solution, 25 mL of methylene blue indicator (10 mL of 0.3 percent methylene blue solution, 8 mL concentrated sulfuric acid, 50 grams sodium sulfate R. diluted to one liter with distilled water), 25 mL of chloroform, and 25 mL of distilled water and the indicated number of drops of sodium arsenite plus an excess of 1 to 2 drops. Titrate with standardized 0.002 N Hyamine 1622 solution in a 50 mL buret. Periodically during titration, shake the graduated cylinder 30 times and observe for color changes in the layers. Continue titration until the color of the layers matches. Record number of milliliters required.

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* 4.4.2.3.1 Calculation. Calculate the percent of active anionic sufactant as follows:

$$\text{Percentage} = \frac{(\text{mL of Hyamine 1622})\ (\text{N Hyamine 1622})\ (341)\ (A)}{(10)\ (B)\ (C)}$$

where A = 250 (the sample dilution)
B = the sample weight
C = 25 (aliquot of sample used)

4.4.2.4 Solubility. Measure into a 1.5 liter beaker, a liter of distilled water at 25° ± 1°C. Weigh out 1 gram of the contents of the trichloromelamine pouch to the nearest mg, add carefully to the distilled water, and stir for 2 minutes with a glass rod. Stirring should not be so rapid as to cause splashing or whipping of the mixture but sufficient to keep the undissolved powder from settling to the bottom. Withdraw the stirring rod and allow the solution to stand 2 minutes. At the end of this time, insert a 20 mL pipette about 1 inch below the surface and draw 20 mL into the pipette. Place this 20 mL sample in a 125 mL Erlenmeyer flask and analyze for titratable chlorine by the following procedure: Dissolve 1 gram of KI crystals in the 20 mL sample. Add 6 drops of 50 percent acetic acid and swirl to ensure complete mixing. Allow to stand 5 minutes in the dark. Titrate tne free iodine at once with freshly standardized 0.005 N sodium thiosulfate. When the color is straw yellow, add starch indicator and titrate to end point.

4.4.2.4.1 Calculation.

$$\text{Milligrams per liter of titratable chlorine} = \frac{(35.45)\ (\text{mL sodium thiosulfate})\ (N)\ (1000)}{(\text{mL of sample solution})}$$

4.4.2.5 Moisture. Weigh to the nearest 0.1 mg a sample of 3 grams of the contents of the trichloromelamine pouch in a covered sample dish. Uncover the dish and dry the sample in a vacuum oven for 2 hours at 50° ± 2°C. Replace the cover, cool the sample to room temperature in a desiccator, and again weigh the sample to the nearest 0.1 mg.

4.4.2.5.1 Calculation.

$$\text{Percent moisture} = \frac{(\text{loss of weight})\ (100)}{(\text{grams of sample})}$$

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4.4.2.6 Particle size. Weigh a representative 10 + 0.05-gram sample of the contents of the trichloromelamine pouch and place on a No. 10 standard sieve conforming to RR-S-366 that is fitted into a No. 20 standard sieve conforming to RR-S-366. The No. 20 sieve is fitted on the receiving pan. After introducing the sample, place a top on the No. 10 sieve. Gently tap the sieve assembly until no further material will pass through either sieve, as determined by observation. Weigh the material that was collected in the bottom pan to the nearest 0.05 gram. The weight of the sample shall be not less than 9.5 grams.

4.4.2.6.1 Calculation.

Percent passing No. 20 sieve = (grams of material in pan) (10)

4.4.2.7 Bactericidal performance. The following procedure should be carried out on three separate 100 mg portions of the contents of the trichloromelamine pouch to give three independent sets of results:

a. Aseptically transfer to four sterilized 250 mL Erlenmeyer flasks 93 mL of cooled, sterilized hard water prepared as follows: To one liter of sterilized, cooled, glass distilled water, add 0.54 gram of sodium bicarbonate and 0.36 grams of anhydrous calcium chloride.

CAUTION: Add these ingredients aseptically only to the cooled sterilized water; hot water may cause decomposition of the bicarbonate.

b. Mark the flasks 1, 2, 3, and 4 and place in a 45° ± 1°C constant temperature water bath for 30 minutes or more. Weigh three 100-mg samples of the finished product of the trichloromelamine pouch to the nearest 0.1 mg and introduce them separately into flasks 1, 2, and 3. Swirl the contents to facilitate solution and return the flasks to the bath.

c. After 10 minutes, pipette 2 mL of a freshly made, sterile 10 percent Difco Bacto-Peptone solution or equal into each of the four flasks. Swirl the contents again and return to the bath.

d. After 120 minutes, pipette 5 mL of a suspension of bacterium E. coli (strain No. 26ATCC) (see 4.4.2.7.1) containing approximately two billion cells per mL into each of the four flasks, using sterile pipettes, and mix thoroughly by continuous swirling in the bath.

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e. After a contact of 90 seconds, pipette 2 mL of the final mixture in each of flasks 1, 2, and 3 separately into three sterile test tubes each containing 2 mL of a sterile 1 percent sodium sulfite solution (made up to contain 1.0 percent Na_2SO_3, 0.1 percent $Na_4P_2O_7 10H_2O$, and 0.2 percent glycerine) that is prepared no more than 24 hours before use. (Note: All glassware used in contact with the test fluid must be sterilized, and a fresh pipette must be used for each sample. Both the bacteria-dosing and the taking of sample must be so timed that the bacteria in each of the three test flasks shall have exactly 90 seconds of contact with the test fluid.)

f. Pour triplicate plates with 1 mL portions of each of the three neutralized samples before counting using tryptone glucose extract agar. Remove a 1 mL portion from flask 4 into 99 mL sterile water and mix thoroughly using tryptone glucose extract agar. Make two more 1:100 serial dilutions in the same manner, and pour triplicate plates with 0.1 and 1.0 mL portions of the $1:10^6$ dilution using tryptone glucose extract agar and incubate at 37° ± 1°C for 48 hours before counting. Multiply by two the counts on plates poured with the three test samples to correct for the dilution. Compute the percentage of survivors in each of the three test flasks from the average count per milliliter of each of the test samples and that of the control sample from flask 4.

4.4.2.7.1 Inoculum. Wash off the surface growth on several 18-hour old nutrient-agar slants with sterile, normal saline, pooled and filtered through sterile No. 2 Whatman filter paper. Adjust the suspension to 2.0 billion cells per milliliter with a photoelectric colorimeter or other suitable means. Store in the refrigerator until used. The suspension must be prepared fresh each day.

4.4.2.8 Formulation stability. Seal triplicate samples of the contents of the trichloromelamine pouch of approximately 50 grams each (the tritratable chlorine having been determined by the method specified in 4.4.2.1) in 4-ounce screwcap glass jars or other suitable glass containers and store for 60 days in a suitable constant-temperature oven maintained at 50° ± 1°C. At the end of the test period, determine the titratable chlorine of each sample by the method specified in 4.4.2.1. The average of the three samples shall be taken as the final value of titratable chlorine.

4.4.2.8.1 Calculation.

$$\text{Percent titratable chlorine remaining} = \frac{(\text{final value})\ (100)}{(\text{original value})}$$

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4.4.2.9 Surface tension. Grind and mix thoroughly approximately 5 grams of the contents of the trichloromelamine pouch in a glass mortar with a glass pestle. Weigh a 1-gram sample to the nearest 0.1 mg, obtained by sampling various portions of this finely ground and well mixed material, and dissolve in distilled water. Make up to 1 liter volume. Three suitable portions of this solution shall be used for determination of surface tension at 25°C using a calibrated du Nouy tensiometer or a Roller Smith surface tension balance (platinum ring method). The average of the three determinations is obtained and is expressed as dynes per centimeter.

4.4.3 Tests on filled and sealed pouches.

4.4.3.1 Finished pouch stability. Place the finished pouch or pouches, as applicable, (filled and heat sealed) in a humidity cabinet for a period of 15 days. Maintain the temperature and humidity range of the test cabinet at 38° ± 1°C with a minimum relative humidity of 90 percent.

4.4.3.2 Stability after packaging. Place the finished pouch or pouches, as applicable (filled and heat sealed), in a test cabinet maintained at 59° ± 1°C for a period of 15 days.

4.4.3.3 Leakage of seams and closures of pouch. Place one filled and heat sealed pouch in a desiccator, or other suitable container, partially filled with water. Cover the pouch with a grating to prevent floating and to maintain a one-inch height of water over the top surface of the filled pouch. Gradually apply a vacuum of 20 inches of mercury for a period of 3 minutes. Observe the pouch for a steady flow of air bubbles (leakage indicating) during this period. Isolated or static bubbles caused by entrapped air in the seams or on the surfaces of the pouch will not be considered as a sign of leaks. Remove the pouch from the desiccator or other suitable container, dry hands and pouch, and examine the inside of the pouch for presence of moisture.

4.4.3.4 Water spray test. Place the completed intermediate pack, filled and sealed as specified in 5.1.1, on a rack 1 inch in height and arrange side by side in the same position as they will be placed in the exterior shipping container. The spray shall consist of a spray of water covering the complete upper surfaces of the four intermediate packs. There shall be a minimum free fall of spray of 5 feet. The average fall of water shall be at the minimum rate of 2 gallons per square foot per hour. The spray shall be continuous for 4 hours.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

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5.1.1 Level A. Twelve bagged double-compartmented pouches of disinfectant, with an instruction sheet (see 3.5.2), shall be packed in a snug-fitting intermediate nontest corrugated fiberboard box; a folding paperboard box conforming to variety 1, style III, type G, class i of PPP-B-566; or a setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. Each fiberboard box shall be closed by means of tape. Each paperboard box shall be closed in accordance with the appendix of the applicable box document. Each box shall then be placed in a waterproof 0.002-inch thick polyethylene bag conforming to type I, grade A, style 1 of PPP-B-26. Heat sealed seams shall be a minimum of 1/16 inch in width. Prior to or during the heat seal closure operation, excess air within the bag shall be expelled. The completed intermediate pack shall be water spray tested as specified in 4.4.3.4. After testing, there shall be no water or soggy or wet material within the intermediate pack.

* 5.1.2 Commercial. Disinfectant, with an instruction sheet (see 3.5.2), shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Ninety-six pouches (eight intermediate packs) of disinfectant, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. Level A intermediate packs shall be packed flat, two in length, two in width, and two in depth within a shipping container. Each wood container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Ninety-six pouches (eight intermediate packs) of disinfectant, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed flat, two in length, two in width, and two in depth within a shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

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* 5.2.3 Commercial packing. Disinfectant, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), disinfectant, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type III, type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Special marking. Each intermediate pack shall have the following legend above the nomenclature:

This Box Contains 12-Each Double-Compartmented Pouches of (Lot Number).

5.4.2 Shelf-life marking. Each intermediate pack and shipping container shall be marked with the following shelf-life information located on same side and/or ends as the identification data in the order listed:

Type II Shelf-Life Item
Date of Manufacture _____
Expiration Date _____
Inspection/Test Date _____ Test No._____
Extended/Reinspection Date _____

5.4.2.1 Additional shelf-life marking. A label shall be affixed in accordance with MIL-STD-129 to each shipping container and the following legend shall be printed in white ink on red paper:

CAUTION

NOT TO BE USED AFTER _____*

UNLESS CONTENTS YIELD A MINIMUM OF

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17 PERCENT TITRATABLE CHLORINE WHEN

TESTED IN ACCORDANCE WITH INSTRUCTIONS

INSERTED IN INTERIOR PACKAGE

* Supplier shall insert the month and year that is 12 months after the date of pack.

6. NOTES

6.1 Intended use. The food service disinfectant is intended for use in disinfection of fresh fruits and vegetables grown in areas where use of feces as fertilizer is suspected due to local customs. The food service disinfectant is also intended for use in disinfection of mess gear and eating utensils. The food service disinfectant provides health protection to personnel against diseases such as amoebic dysentery, and under emergency conditions, it constitutes their sole protection. It is intended for use by troops, and the amount packaged is sufficient for treating 25 gallons of water and 100 mess kits. The food service disinfectant covered by this document is intended for use only when hot water rinse is not available.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When laboratory report is not required (see 3.1).
c. When weight of pouch required differs from specified weight (see 3.3.1).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Component handling. Extreme care must be exercised in handling, grinding, mixing, and packaging the components and formulation to protect against moisture and to ensure homogeneity of the formulation so that a consistently high quality product will be produced that will remain stable during long time storage. All equipment should be kept clean and protected against moisture during any shutdown. During a shutdown of more than 12 hours, it is recommended that all adhering formulation be removed from the grinding, mixing, and filling machines.

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6.4 Specialized requirements. Contractor's attention is directed to the fact that this item is most difficult to produce and requires specialized equipment and technical know-how based on past experience to produce the superior product required. There can be no compromise or deviation from the high quality and stability standards set for this product. Components, all but for surface active material and potassium iodide, having the properties listed below have been used successfully in the past to prepare a food service disinfectant that conforms to this document.

6.4.1 Trichloromelamine. The trichloromelamine should be white to light cream in color, having a titratable chlorine of not less than 89.0 percent and no more than 94.5 percent, and a moisture content not greater than 0.5 percent.

6.4.2 Citric acid, anhydrous. The anhydrous citric acid should conform to the requirements of U.S. Pharmacopeia, have a moisture content of not greater than 0.2 percent, and the assay show not less than 99.3 percent $C_6H_8O_7$.

6.4.3 Monosodium dihydrogen phosphate, anhydrous. The anhydrous monosodium dihydrogen phosphate should contain no less than 57.5 percent nor more than 59.1 percent of phosphorous pentoxide and should have a moisture content no greater than 2.0 percent.

6.5 Grinding. In order to meet solubility requirements consistently, it is necessary to grind at least a portion of the finished formulation and all of the trichloromelamine and wetting agent in micronizing equipment. *Trichloromelamine should not be ground alone or when mixed only with the* monosodium dihydrogen phosphate because of fire hazard.

6.6 Changes to previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Air Force - 68

Review activities:

Army - MD
Navy - MC
Air Force - 03
DLA - GS

User activity:

Navy - MS

Preparing activity:

Army - GL

Project No. 6840-0371

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DISINFECTANT, FOOD SERVICE
(CHLORINE-IODINE TYPE)
6840-00-810-6396
MIL-D-11309

For disinfection of mess gear, fresh fruits and vegetables

(4.77 oz. net quantity for 25 gallons of water)

ACTIVE INGREDIENTS:

Trichloromelamine	**19.3%**
Potassium Iodide	**28.6%**
INERT INGREDIENTS:	**52.1%**

AVAILABLE CHLORINE 250 mg/l (ppm)

TITRATABLE IODINE 300 mg/l (ppm)

DANGER

SEE BACK PANEL FOR ADDITIONAL PRECAUTIONS AND FIRST AID

Name of Mfr. & Address **Lot No.** **Date**

Contract No.

EPA Reg. No. 40510-1 **EPA Est. No.**

Inspection / Test Date

Figure 1. EPA Approved Front Label

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DIRECTIONS FOR USE

IT IS A VIOLATION OF FEDERAL LAW TO USE THIS PRODUCT IN A MANNER INCONSISTENT WITH ITS LABELING

a. For disinfection of mess gear (intended for use only when hot water rinse is not available):

The double-compartmented pouch is sufficient for treating 25 gallons of water resulting in a dosage of approximately 257 mg/l of chlorine and 300 mg/l of iodine. To treat 100 sets of mess gear, dissolve the contents of one double compartmented pouch of Disinfectant, Food Service (consisting of one trichloromelamine pouch labeled pouch A and one potassium iodide pouch labeled pouch B) in a container containing 25 gallons of water. Repeat the above procedure with a second container containing 25 gallons of rinse water. If water is cold, mix contents of the double-compartmented pouch with 3 to 4 parts of water in a small container (canteen cup will do) and pour the mixture into the rinse water. Repeat for the second container of 25 gallons of water. Stir thoroughly to dissolve. After washing each set of mess gear, rinse twice-once in the first container of disinfectant solution and a final rinse in the second container of disinfectant solution. Allow mess gear to air dry. A new solution must be prepared after disinfecting 100 mess kits.

b. For fresh fruits and vegetables:

1. Remove bruised and torn leaves and damaged fruits or vegetables.
2. Do not cut or peel fruits or vegetables before disinfecting.
3. Wash thoroughly in a solution made by dissolving the contents of one double-compartmented pouch (pouch A and pouch B) in 20 gallons of water. This solution will yield approximately 320 mg/l of chlorine and 380 mg/l of iodine.
4. After washing, completely immerse the fruits and vegetables for ten (10) minutes in a separate freshly prepared solution made by dissolving one double-compartmented pouch of Disinfectant, Food Service in 20 gallons of water. Stir occasionally to obtain thorough wetting of vegetables and fruit surfaces.
5. After ten (10) minutes of immersion, remove the fruits and vegetables from the disinfectant solution and rinse thoroughly in potable water.

NOTE: SOLUTIONS OF DISINFECTANT, FOOD SERVICE ARE NOT TO BE REUSED. MAKE FRESH SOLUTIONS FOR EACH USE.

STORAGE AND DISPOSAL

Store in a cool, dry, well ventilated area away from contact with other chemicals and solvents. Moisture contamination may liberate irritating and hazardous chlorine containing gases. Rinse empty pouches thoroughly with water and discard them in a safe place.

PRECAUTIONARY STATEMENTS HAZARDOUS TO HUMANS AND DOMESTIC ANIMALS

Harmful if swallowed. May cause eye damage. Do not get in eyes, on skin or clothing. Do not breathe dust.

If swallowed: Do not use gastric lavage or emesis. Dilute the chlorine/iodine disinfectant by drinking large quantities of water or milk. Transport to a physician. In case of personal contact: Remove excess material and flush skin and eyes with fresh water for at least 15 minutes. For eyes get medical attention.

PHYSICAL OR CHEMICAL HAZARD

Contains trichloromelamine which may react vigorously to evolve heat and highly toxic smoke under the following conditions: If it is exposed to extreme heat, if ignited by spark or flame, if mixed before dilution in water with certain organic materials, acids and alkalis.

DO NOT MIX WITH OTHER CHEMICALS SUCH AS CLEANING COMPOUNDS, AMMONIA, ACIDS, SOLVENTS, OTHER ORGANIC MATERIAL.

Figure 2. EPA Approved Back Label

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-11309F	Disinfectant, Food Service

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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